

Threshold Entry in a Fixed-Prize Contest

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1 Introduction

Table 1: Headline claims, each traced to the model rows it rests on.

ID	Headline	Model rows	Scope note
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2 Model

Table 2: Results of the model, each with its statement and the Lean theorems that check it.

ID	Claim	Statement	Lean	Proof
M1	Representation (P1).	$\Delta(m) = V \int \varphi dH_m = V \mathbb{E}[\varphi(M_m)]$, with $M_m \sim H_m$.	EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation	p. 5
M2	Dominance (P2).	For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q, u]$ with $q < u$.	EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives	p. 5
M3	Step identity (P3).	For $m+2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \varphi^2 dK_m$.	EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step	p. 6
M4	Monotonicity.	$\Delta(m+1) \leq \Delta(m)$ for $V \geq 0$, with $<$ when $V > 0$ and $\int \varphi^2 dK_m > 0$.	EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict	p. 6
M5	Threshold entry (P5).	For $\kappa_0 \leq \dots \leq \kappa_{Q-1}$ and Δ non-increasing, $\kappa_j \leq \Delta(j)$ and $i \leq j$ give $\kappa_i \leq \Delta(i)$.	EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives	p. 6
M6	Count invariance.	Every equilibrium S , with $\kappa_i \leq \Delta(S - 1)$ for $i \in S$ and $\Delta(S) < \kappa_j$ for $j \notin S$, has $ S = k^*$.	EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le	p. 6

ID	Claim	Statement	Lean	Proof
M7	Identities pinned.	If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \dots, k^* - 1\}$ is the only equilibrium.	EntryContest.members_ below_kstar, EntryContestEq. assortative_unique	p. 7
M8	Assortative selection (N4).	$\{0, \dots, k^* - 1\}$ is an equilibrium; $\sum_{i < k} \kappa_i \leq \sum_{i \in S} \kappa_i$ for $ S = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums.	EntryContestEq. assortative_is_ equilibrium, EntryContestEq. assortative_min_cost, EntryContestEq.prize_ term_anonymous, EntryContestEq.payoff_ gap_anonymous	p. 7
M9	Benchmark (P6).	With the incumbent investing and $s \geq 0$ atomless, $\Delta(m) = V/(m+2)$ at $\mu = 0$ for every Q , and $\Delta(m) \rightarrow V/(m+2)$ as $\mu \rightarrow 0$.	EntryContestP6P7. integral_iid, EntryContestP6P7. integral_cdf_pow, EntryContestP6P7.p6_ evaluation, EntryContestP6P7.p6_ at_zero, EntryContestP6P7. tendsto_cdf_ investorLaw, EntryContestP6P7. tendsto_cdf_ nonInvestorLaw, EntryContestP6P7.p6_ limit	p. 7
M10	Saturation (P7).	$\Delta(m) \leq V/(m+1)$ for every Q , and an entrant at index m has $(m+1)\kappa \leq V$.	EntryContestP6P7. integral_cdf_pow, EntryContestP6P7.p7_ Delta_le, EntryContestP6P7.p7_ cap, EntryContestP6P7. p7_count	p. 8
M11	Prize and fee (P8).	k^* is non-decreasing in V and, for u non-decreasing, non-increasing in c .	EntryContestEq.kstar_ mono, EntryContestEq. Delta_mono_prize, EntryContestMM.Delta_ scale, EntryContestEq. kappa_mono_fee	p. 8
M12	Margin condition.	For sorted profiles w and w' , $w_{k^*-1} \leq w'_{k^*-1}$ gives $k^* \leq k'^*$, and $w'_{k^*} \leq w_{k^*}$ gives $k'^* \leq k^*$.	EntryContestSpread. enters_down, EntryContestSpread. count_ge_of_enters, EntryContestSpread. count_le_of_fails, EntryContestSpread. margin_rise, EntryContestSpread. margin_fall	p. 8
M13	Pivot spreads (P9-gen, P9).	For a pivot-spread T about x_0 , k^* weakly rises when $w_{k^*-1} \geq x_0$ and weakly falls when $w_{k^*-1} \leq x_0$. The linear spread about the mean is one, and keeps the mean. A change that keeps the total and majorizes w can lower k^* with w_{k^*-1} above the mean.	EntryContestSpread. pivot_rise, EntryContestSpread. pivot_fall, EntryContestSpread. spread_isPivot, EntryContestMM.mm2a, EntryContestSpread. mps_lowers_count	p. 8

ID	Claim	Statement	Lean	Proof
M14	The band.	With $d = w' - w$, the signs of d leave the direction of k^* open only when $d_{k^*-1} < 0 < d_{k^*}$. A pivot-spread never does so. Inside the band two changes that keep the total and majorize w , with the same signs at every rank, move k^* in opposite directions for every small $\mu > 0$.	EntryContestSpread. branch_of_not_band, EntryContestSpread. not_band_of_single_crossing, EntryContestSpread. pivot_single_crossing, EntryContestSpread. mps_band_signs, EntryContestSpread. mps_witness_counts, EntryContestSpread. mps_band_both_directions, EntryContestSpread. mps_band_both_directions_positive	p. 9
M15	Strictness under a spread (P9).	If some entrant lies below x_0 at $\lambda = 1$ and $\kappa(w) \rightarrow \infty$ as $w \downarrow c$, $k^*(\lambda)$ is non-increasing in $\lambda \geq 1$ and $k^*(\lambda) < k^*(1)$ above some $\bar{\lambda}$.	EntryContestKappa. exit_below_pivot, EntryContestKappa. enters_downward, EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone, EntryContestKappa.p9_strict_model	p. 9
M16	Competitor number (P-MU).	For $Q \geq 1$ and every C , $D(Q) = -V(\int W_Q dF - \int W_Q dG)$, so for $V > 0$, $D(Q) > 0$ exactly when $\int W_Q dF < \int W_Q dG$. The kernel $g^n(1 - g)$ rises on $[0, \frac{n}{n+1}]$ and falls on $[\frac{n}{n+1}, 1]$.	EntryContestPMU. weight_eq, EntryContestPMU.pmu_identity, EntryContestPMU.pmu_sign_iff, EntryContestPMU. kernel_rises, EntryContestPMU. kernel_falls, EntryContestPMU. kernel_max	p. 10
M17	Rise then fall in Q .	If $dF - dG \leq 0$ on $(-\infty, x_0]$ and $dF - dG \geq 0$ on (x_0, ∞) , and $V \geq 0$, then $D(Q_2) \leq G(x_0)^{Q_2 - Q_1} D(Q_1)$ for $1 \leq Q_1 \leq Q_2$, and $\min(\Delta_{Q_1}(0), \Delta_{Q_3}(0)) \leq \Delta_{Q_2}(0)$ for $1 \leq Q_1 \leq Q_2 \leq Q_3$.	EntryContestPMU. weight_shift, EntryContestPMU.pmu_single_crossing, EntryContestPMU.pmu_step_nonpos_persists, EntryContestPMU. quasiconcave_of_steps, EntryContestPMU.pmu_quasiconcave	p. 10
M18	Uniform base score.	For r uniform on $[0, 1]$, $s \geq 0$, $0 < \mu \leq 1$, $V \geq 0$ and every C , the hypothesis of M17 holds at $x_0 = \mu$, and $D(Q) \geq 0$ for $Q \geq 1$, so $\Delta_Q(0)$ is non-decreasing in $Q \geq 1$.	EntryContestPMU. uniform_left, EntryContestPMU. uniform_right, EntryContestPMU.pmu_quasiconcave_uniform, EntryContestRefute. pmu_step_nonneg_uniform, EntryContestRefute. Delta_mono_uniform	p. 10

ID	Claim	Statement	Lean	Proof
M19	A class with rise then fall.	For $s \in \{0, 1\}$ with $\Pr(s = 0) = p$, $0 < \mu < 1$, $V \geq 0$, $\Pr(r \leq 0) = 0$ and $\Pr(r \in B) \leq \Pr(r \in B - c)$ whenever $c > 0$ and $B \subseteq (c, \infty)$, the hypothesis of M17 holds at $x_0 = 1 - \mu$, and $\Delta_Q(0)$ is quasi-concave in $Q \geq 1$ for every C . A density of r non-increasing on $(0, \infty)$ gives the shift condition, and the smaller of two uniform draws has one.	EntryContestShift. left_of_lift, EntryContestShift. right_of_lift, EntryContestShift. quasiconcave_of_ shiftMono, EntryContestShift. shiftMono_of_density, EntryContestShift. lawA_shiftMono, EntryContestShift. witness_quasiconcave	p. 11
M20	The fall, exhibited.	For r the smaller of two uniform draws, $\mu = \frac{3}{4}$, s as in M19 and $C = F$, $D(1)$ and $D(2)$ are quadratics in p (in the proof). For $V > 0$, $p = \frac{1}{2}$ gives $\Delta_1(0) < \Delta_2(0)$ and $\Delta_3(0) < \Delta_2(0)$, and $p = \frac{1}{4}$ gives $\Delta_2(0) < \Delta_1(0)$.	EntryContestFall. stepQ_one, EntryContestFall. stepQ_two, EntryContestFall.rise_ then_fall, EntryContestFall.fall_ from_one	p. 11
M21	When the gain must fall.	With s and r as in M19, $t = 1 - \mu$ and G_t the law of $\mu r + t$, if $t < a < b$, $G(t) < G(a)$, $G(b) < 1$, $C(a) > 0$ and $G((a, b]) < G_t((a, b])$, then for $V > 0$ and $p < 1$ some Q_0 has $D(Q) < 0$ for every $Q \geq Q_0$. The base law of M20 meets this on $(\frac{1}{2}, \frac{5}{8}]$ for every $p < 1$, and a uniform base score never does.	EntryContestMustFall. must_fall_general, EntryContestMustFall. shift_dominates, EntryContestMustFall. stepQ_bern, EntryContestMustFall. must_fall, EntryContestMustFall. witness_must_fall, EntryContestMustFall. unif_shiftMono, EntryContestMustFall. uniform_never_gains	p. 11
M22	Dominance does not sign $D(Q)$.	With $C = F$, the pairs $(F, G) = (x, 1 - (1 - x)^2)$ and (x^2, x) on $[0, 1]$ both have $F \leq G$ and no atoms. The first has $D(1) = \frac{V}{60}$ and $D(2) = \frac{V}{420}$, and the second has $D(Q) = -\frac{VQ}{(Q+2)(Q+3)(Q+4)}$.	EntryContestPMUWitness. pairA_fosd, EntryContestPMUWitness. pairB_fosd, EntryContestPMUWitness. pairA_step_one, EntryContestPMUWitness. pairA_step_two, EntryContestPMUWitness. pairB_step, EntryContestPMUWitness. fosd_does_not_sign	p. 12

3 Literature

Table 3: Papers the model or the conclusions depend on, each with a verbatim quote.

ID	Paper	What we rely on	Quote	Page	Model rows
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4 Conclusions

Table 4: The novel claims, each linked to the model and literature rows it rests on.

ID	Statement	Rests on
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A Proofs

Notation. Challenger i scores $\sigma_i = \mu r_i + (1 - \mu)s_i e_i$, with r_i and $s_i \geq 0$ drawn independently of wealth. F and G are the score CDFs of an entrant and of an outsider, and C is the incumbent's, each without atoms. With m other entrants among the Q challengers,

$$H_m = C F^m G^{Q-1-m}, \quad K_m = C F^m G^{Q-2-m}, \quad \phi = G - F,$$

$$\Delta(m) = V \left(\int H_m dF - \int H_m dG \right), \quad \kappa(w) = u(w) - u(w - c).$$

Challengers are indexed $0, \dots, Q - 1$ in increasing order of cost, $\kappa_0 \leq \dots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index j enters when $\kappa_j \leq \Delta(j)$, and k^* satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$. In the results on wealth profiles, rank j has wealth w_j with $w_0 \geq w_1 \geq \dots$, and enters when $w_j > c$ and $\kappa(w_j) \leq \Delta(j)$, so a challenger whose wealth is at or below the fee does not enter. A prime marks the profile after a spread and its count k^* . When the number of challengers varies, $\Delta_Q(0)$ is the first entrant's gain among Q challengers, with

$$D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0), \quad W_Q = C G^{Q-1}(1 - G).$$

Proof of M1

Lean: EntryContestP1P2.two_max_sum, EntryContestP1P2.p1_representation, EntryContestP1P2.Delta_eq_expectation.

For independent $X \sim A$ and $Y \sim B$ with B atomless, $\Pr(X = Y) = 0$, so

$$\int B dA + \int A dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A, B) = (F, H_m)$ and $(A, B) = (G, H_m)$,

$$\int H_m dF = 1 - \int F dH_m, \quad \int H_m dG = 1 - \int G dH_m,$$

so

$$\frac{\Delta(m)}{V} = \int (G - F) dH_m = \int \phi dH_m = \mathbb{E}[\phi(M_m)].$$

□

Proof of M2

Lean: EntryContestP1P2.p2_fosd, EntryContestP1P2.p2_strict_interval, EntryContestP1P2.Delta_nonneg_from_primitives.

Let $X = \mu r + (1 - \mu)s$ and $Y = \mu r$. For $\mu \leq 1$ and $s \geq 0$, $X - Y = (1 - \mu)s \geq 0$, so $\{X \leq x\} \subseteq \{Y \leq x\} \cup \{s < 0\}$ and

$$F(x) \leq G(x) + \Pr(s < 0) = G(x), \quad \Delta(m) = V \int \phi dH_m \geq 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \leq q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \leq q < X) > 0$, and

$$G(q) \geq \Pr(X \leq q) + \Pr(Y \leq q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q, u)$.

□

Proof of M3

Lean: EntryContestStep.three_max_sum_real, EntryContestStep.step_identity, EntryContestModel.Delta_step.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K, X \sim A, Y \sim B$, exactly one is the largest,

$$\int AB dK + \int KB dA + \int KA dB = 1.$$

With $(A, B) = (F, F), (G, G)$ and (F, G) ,

$$\int KF dF = \frac{1}{2} \left(1 - \int F^2 dK \right), \quad \int KG dG = \frac{1}{2} \left(1 - \int G^2 dK \right), \quad \int KG dF + \int KF dG = 1 - \int FG dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF dF - \int KF dG - \int KG dF + \int KG dG = -\frac{1}{2} \int (F - G)^2 dK.$$

□

Proof of M4

Lean: EntryContestModel.Delta_step_nonpos, EntryContestModel.DeltaUpTo_step, EntryContestEq.Delta_step_strict.

By M3 and $\phi^2 \geq 0$,

$$\Delta(m+1) - \Delta(m) = -\frac{V}{2} \int \phi^2 dK_m \leq 0 \quad (V \geq 0), \quad < 0 \quad \left(V > 0, \int \phi^2 dK_m > 0 \right).$$

For $m \geq Q - 1$ the gain is extended as $\Delta(Q - 1)$, so $\Delta(m+1) \leq \Delta(m)$ for every m .

□

Proof of M5

Lean: EntryContest.single_crossing, EntryContestModel.p5_threshold_from_primitives.

For $i \leq j$, by M4,

$$\kappa_i \leq \kappa_j \leq \Delta(j) \leq \Delta(i).$$

□

Proof of M6

Lean: EntryContestEq.count_invariance_fin, EntryContestMM.exists_mem_ge, EntryContestMM.exists_not_mem_le.

Let $k = |S|$. A k -set of indices has an element $i \geq k - 1$, since $S \subseteq \{0, \dots, k - 2\}$ gives $k \leq k - 1$, and omits some $j \leq k$, since $\{0, \dots, k\} \subseteq S$ gives $k + 1 \leq k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \leq \kappa_{k-1} \leq \kappa_i \leq \Delta(k - 1) \leq \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \leq \kappa_k \leq \Delta(k).$$

Both are contradictions, so $k = k^*$.

□

Proof of M7

Lean: EntryContest.members_below_kstar, EntryContestEq.assortative_unique.

A member i of an equilibrium has $\kappa_i \leq \Delta(k^* - 1)$ by M6. If $i \geq k^*$,

$$\kappa_i \geq \kappa_{k^*} > \Delta(k^* - 1),$$

so $i < k^*$. Then $S \subseteq \{0, \dots, k^* - 1\}$ and $|S| = k^*$ give $S = \{0, \dots, k^* - 1\}$. □

Proof of M8

Lean: EntryContestEq.assortative_is_equilibrium, EntryContestEq.assortative_min_cost, EntryContestEq.prize_term_anonymous, EntryContestEq.payoff_gap_anonymous.

(i) For $i < k^*$ and for $k^* \leq j < Q$,

$$\kappa_i \leq \kappa_{k^*-1} \leq \Delta(k^* - 1), \quad \Delta(k^*) < \kappa_{k^*} \leq \kappa_j.$$

(ii) Induct on S , adding its largest element a last. With $T = S \setminus \{a\}$ and $T \subseteq \{0, \dots, a-1\}$, $|T| \leq a$, so

$$\sum_{i < |T|+1} \kappa_i = \sum_{i < |T|} \kappa_i + \kappa_{|T|} \leq \sum_{i \in T} \kappa_i + \kappa_a = \sum_{i \in S} \kappa_i.$$

(iii) With identical laws, the win probability of i is $P^{\text{in}}(|S|) = \int CF^{|S|-1} G^{Q-|S|} dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|} G^{Q-1-|S|} dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i \in S} \kappa_i + V(|S|P^{\text{in}}(|S|) + (Q - |S|)P^{\text{out}}(|S|)),$$

and the last term is equal for $|S| = |T|$. □

Proof of M9

Lean: EntryContestP6P7.integral_iid, EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p6_evaluation, EntryContestP6P7.p6_at_zero, EntryContestP6P7.tendsto_cdf_investorLaw, EntryContestP6P7.tendsto_cdf_nonInvestorLaw, EntryContestP6P7.p6_limit.

For $X_0, \dots, X_k$ independent with atomless CDF F ,

$$\int F^k dF = \Pr\left(X_0 = \max_{i \leq k} X_i\right) = \frac{1}{k+1}.$$

At $\mu = 0$ the entrant scores s and the outsider 0, so $G = \mathbf{1}_{[0, \infty)}$ and $F(0) = \Pr(s \leq 0) = 0$. With $C = F$,

$$\int H_m dF = \int F^{m+1} dF = \frac{1}{m+2}, \quad \int H_m dG = H_m(0) = 0.$$

For $\mu \rightarrow 0$ let $X_\mu = \mu r + (1 - \mu)s$ and $Y_\mu = \mu r$. For $s > 0$,

$$F_\mu(X_\mu) \rightarrow F_0(s), \quad G_\mu(X_\mu) \rightarrow 1, \quad F_\mu(Y_\mu) \rightarrow F_0(0) = 0,$$

with every integrand in $[0, 1]$, so by dominated convergence

$$\int H_m dF_\mu \rightarrow \int F_0^{m+1} dF_0 = \frac{1}{m+2}, \quad 0 \leq \int H_m dG_\mu \leq \mathbb{E}[F_\mu(Y_\mu)] \rightarrow 0.$$

□

Proof of M10

Lean: EntryContestP6P7.integral_cdf_pow, EntryContestP6P7.p7_Delta_le, EntryContestP6P7.p7_cap, EntryContestP6P7.p7_count.

From $C \leq 1$, $G \leq 1$ and $\int H_m dG \geq 0$,

$$\Delta(m) \leq V \int H_m dF \leq V \int F^m dF = \frac{V}{m+1}.$$

An entrant at index m has $\kappa \leq \Delta(m)$, so

$$(m+1)\kappa \leq (m+1)\Delta(m) \leq V, \quad m+1 \leq \frac{V}{\kappa} \quad (\kappa > 0).$$

□

Proof of M11

Lean: EntryContestEq.kstar_mono, EntryContestEq.Delta_mono_prize, EntryContestMM.Delta_scale, EntryContestEq.kappa_mono_fee.

Let k^* and k'^* be the counts for (κ, Δ) and (κ', Δ') , with $\kappa'_j \leq \kappa_j$ and $\Delta(j) \leq \Delta'(j)$ for every j . If $k'^* < k^*$, then $k'^* < Q$ and

$$\Delta'(k'^*) < \kappa'_{k'^*} \leq \kappa_{k'^*} \leq \Delta(k'^*) \leq \Delta'(k'^*),$$

so $k^* \leq k'^*$. By M1 and M2, $\Delta_V(j) = V \Delta_1(j)$ with $\Delta_1(j) \geq 0$, and for u non-decreasing,

$$V \leq V' \Rightarrow \Delta_V \leq \Delta_{V'}, \quad c \leq c' \Rightarrow u(w) - u(w-c) \leq u(w) - u(w-c').$$

□

Proof of M12

Lean:

EntryContestSpread.enters_down, EntryContestSpread.count_ge_of_enters, EntryContestSpread.count_le_of_fails, EntryContestSpread.margin_rise, EntryContestSpread.margin_fall.

(i) The marginal entrant has $c < w_{k^*-1} \leq w'_{k^*-1}$, so

$$\kappa(w'_{k^*-1}) \leq \kappa(w_{k^*-1}) \leq \Delta(k^* - 1),$$

and for $i \leq k^* - 1$, with $w'_i \geq w'_{k^*-1} > c$,

$$\kappa(w'_i) \leq \kappa(w'_{k^*-1}) \leq \Delta(k^* - 1) \leq \Delta(i).$$

Every $i < k^*$ enters after the spread, so $k^* \leq k'^*$. (ii) For $k^* < Q$, rank k^* fails before the spread. If $w'_{k^*} > c$, then $c < w'_{k^*} \leq w_{k^*}$ and

$$\kappa(w'_{k^*}) \geq \kappa(w_{k^*}) > \Delta(k^*),$$

so rank k^* fails after the spread and $k'^* \leq k^*$.

□

Proof of M13

Lean: EntryContestSpread.pivot_rise, EntryContestSpread.pivot_fall, EntryContestSpread.spread_isPivot, EntryContestMM.mm2a, EntryContestSpread.mps_lowers_count.

T is non-decreasing, so $w'_j = T(w_j)$ is sorted, and

$$x_0 \leq w_{k^*-1} \Rightarrow w_{k^*-1} \leq T(w_{k^*-1}), \quad w_{k^*} \leq w_{k^*-1} \leq x_0 \Rightarrow T(w_{k^*}) \leq w_{k^*},$$

so M12 applies. For $T(w) = x_0 + \lambda(w - x_0)$ with $\lambda \geq 1$,

$$T(w) - w = (\lambda - 1)(w - x_0), \quad \frac{1}{Q} \sum_j T(w_j) = \bar{w} + \lambda \left(\frac{1}{Q} \sum_j w_j - \bar{w} \right) = \bar{w} \quad (x_0 = \bar{w}).$$

With $c = 1$, $\kappa(w) = \frac{1}{w-1} - \frac{1}{w}$ and $\Delta(m) = \frac{1}{m+2}$,

$$w = \left(\frac{29}{10}, \frac{14}{5}, 2 \right), \quad w' = \left(\frac{18}{5}, \frac{11}{5}, \frac{19}{10} \right), \quad \sum_j w_j = \sum_j w'_j, \quad w'_2 \leq w_2, \quad w'_2 + w'_1 \leq w_2 + w_1,$$

$$\kappa(w) = \left(\frac{100}{551}, \frac{25}{126}, \frac{1}{2} \right), \quad \kappa(w') = \left(\frac{25}{234}, \frac{25}{66}, \frac{100}{171} \right), \quad \frac{1}{3} \sum_j w_j = \frac{77}{30} < w_1,$$

so the counts are 2 and 1. □

Proof of M14

Lean: EntryContestSpread.branch_of_not_band, EntryContestSpread.not_band_of_single_crossing, EntryContestSpread.pivot_single_crossing, EntryContestSpread.mps_band_signs, EntryContestSpread.mps_witness_counts, EntryContestSpread.mps_band_both_directions, EntryContestSpread.mps_band_both_directions_positive.

Outside the band $d_{k^*-1} \geq 0$ or $d_{k^*} \leq 0$, and M12 signs the change. If

$$i \leq j, d_j > 0 \Rightarrow d_i \geq 0,$$

then $d_{k^*} > 0$ gives $d_{k^*-1} \geq 0$. A pivot-spread has this property, since $d_j > 0$ gives $w_j > x_0$ and then $w_i \geq w_j > x_0$. With κ and Δ as in M13, the profiles

$$w = \left(4, \frac{14}{5}, \frac{11}{5}, 2 \right), \quad w' = \left(\frac{41}{10}, \frac{27}{10}, \frac{13}{5}, \frac{8}{5} \right), \quad w'' = \left(\frac{23}{5}, \frac{9}{4}, \frac{9}{4}, \frac{19}{10} \right)$$

all sum to 11, with partial sums from the poorest up

$$\left(2, \frac{21}{5}, 7 \right), \quad \left(\frac{8}{5}, \frac{21}{5}, \frac{69}{10} \right), \quad \left(\frac{19}{10}, \frac{83}{20}, \frac{32}{5} \right),$$

and

$$\kappa(w) = \left(\frac{1}{12}, \frac{25}{126}, \frac{25}{66}, \frac{1}{2} \right), \quad \kappa(w') = \left(\frac{100}{1271}, \frac{100}{459}, \frac{25}{104}, \frac{25}{24} \right), \quad \kappa(w'') = \left(\frac{25}{414}, \frac{16}{45}, \frac{16}{45}, \frac{100}{171} \right)$$

against $\Delta = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5} \right)$. The counts are 2, 3 and 1, and d' and d'' both have signs $(+, -, +, -)$. Every inequality is strict, so by M9 the counts hold for every small $\mu > 0$. □

Proof of M15

Lean:

EntryContestKappa.exit_below_pivot, EntryContestKappa.enters_downward, EntryContestKappa.p9_strict, EntryContestKappa.p9_monotone, EntryContestKappa.p9_strict_model.

Let j be an entrant at $\lambda = 1$ with $w_j < x_0$. Since $\kappa(v) \rightarrow \infty$ as $v \downarrow c$, some $u > c$ has $\kappa > \Delta(j)$ on (c, u) , and

$$\lambda > \bar{\lambda} = \max \left(1, \frac{x_0 - u}{x_0 - w_j} \right) \Rightarrow x_0 + \lambda(w_j - x_0) < u,$$

so rank j fails at λ . By the prefix property no rank from j on enters, and

$$k^*(\lambda) \leq j < j+1 \leq k^*(1).$$

For $1 \leq \lambda_1 \leq \lambda_2$, ranks with $w_i \geq x_0$ have $i < j$ and enter at λ_1 , while ranks with $w_i < x_0$ have $x_0 + \lambda_2(w_i - x_0) \leq x_0 + \lambda_1(w_i - x_0)$, so every entrant at λ_2 enters at λ_1 . □

Proof of M16

Lean:

`EntryContestPMU.weight_eq, EntryContestPMU.pmu_identity, EntryContestPMU.pmu_sign_iff, EntryContestPMU.kernel_rises, EntryContestPMU.kernel_falls, EntryContestPMU.kernel_max.`

With no other entrant, $H_0 = CG^{Q-1}$ among Q challengers and CG^Q among $Q+1$, and $CG^{Q-1} - CG^Q = W_Q$, so

$$\frac{D(Q)}{V} = \int (CG^Q - CG^{Q-1}) dF - \int (CG^Q - CG^{Q-1}) dG = - \left(\int W_Q dF - \int W_Q dG \right).$$

For the kernel,

$$\frac{d}{dg} g^n (1-g) = g^{n-1} (n - (n+1)g) \quad (n \geq 1), \quad \frac{d}{dg} (1-g) = -1 \quad (n=0).$$

□

Proof of M17

Lean: `EntryContestPMU.weight_shift, EntryContestPMU.pmu_single_crossing, EntryContestPMU.pmu_step_nonpos_persists, EntryContestPMU.quasiconcave_of_steps, EntryContestPMU.pmu_quasiconcave.`

Let $k = Q_2 - Q_1$, $\rho = G(x_0)^k$ and $v = dF - dG$. By M16 and $W_{Q_2} = G^k W_{Q_1}$,

$$D(Q_2) - \rho D(Q_1) = -V \int (G^k - \rho) W_{Q_1} dv.$$

G is non-decreasing and $W_{Q_1} \geq 0$, so

$$(G^k - \rho) W_{Q_1} \leq 0, \quad v \leq 0 \text{ on } (-\infty, x_0], \quad (G^k - \rho) W_{Q_1} \geq 0, \quad v \geq 0 \text{ on } (x_0, \infty),$$

and $D(Q_2) \leq \rho D(Q_1)$. Then $D(i) \leq 0$ gives $D(j) \leq 0$ for $j \geq i$. If $D(i) \leq 0$ for some $Q_1 \leq i < Q_2$, then $\Delta_{Q_3}(0) \leq \Delta_{Q_2}(0)$, and otherwise $\Delta_{Q_1}(0) \leq \Delta_{Q_2}(0)$.

□

Proof of M18

Lean: `EntryContestPMU.uniform_left, EntryContestPMU.uniform_right, EntryContestPMU.pmu_quasiconcave_uniform, EntryContestRefute.pmu_step_nonneg_uniform, EntryContestRefute.Delta_mono_uniform.`

G is the law of μr , so $G = 1$ on $[\mu, \infty)$, $dG = 0$ on (μ, ∞) and $W_Q = 0$ there. For $B \subseteq (-\infty, \mu]$ and $y \geq 0$, with $\tau = (1 - \mu)y/\mu$ and λ Lebesgue measure,

$$\Pr(\mu r + (1 - \mu)y \in B) = \lambda\left([\tau, 1] \cap \frac{B}{\mu}\right) \leq \lambda\left([0, 1] \cap \frac{B}{\mu}\right) = \Pr(\mu r \in B),$$

so $dF \leq dG$ on $(-\infty, \mu]$ and M17 applies at $x_0 = \mu$. By M16,

$$\frac{D(Q)}{V} = \int_{(-\infty, \mu]} W_Q d(G - F) \geq 0.$$

□

Proof of M19

Lean: EntryContestShift.left_of_lift, EntryContestShift.right_of_lift, EntryContestShift.quasiconcave_of_shiftMono, EntryContestShift.shiftMono_of_density, EntryContestShift.lawA_shiftMono, EntryContestShift.witness_quasiconcave.

Let $t = 1 - \mu$ and G_t be the law of $\mu r + t$, so $F = pG + (1 - p)G_t$. From $\Pr(r \leq 0) = 0$, G_t has no mass on $(-\infty, t]$, where $dF = p dG \leq dG$. For $B \subseteq (t, \infty)$, with $c = t/\mu$ and $B/\mu \subseteq (c, \infty)$,

$$G(B) = \Pr(r \in \frac{B}{\mu}) \leq \Pr(r \in \frac{B}{\mu} - c) = G_t(B),$$

so $dF \geq dG$ on (t, ∞) and M17 applies at $x_0 = t$. For a density g non-increasing on $(0, \infty)$ and $B \subseteq (c, \infty)$,

$$\int_B g(x) dx \leq \int_B g(x - c) dx = \int_{B-c} g(x) dx,$$

and the smaller of two uniform draws has $\Pr(r \leq x) = 1 - (1 - x)^2$ and density $2(1 - x)$ on $[0, 1]$. □

Proof of M20

Lean: EntryContestFall.stepQ_one, EntryContestFall.stepQ_two, EntryContestFall.rise_then_fall, EntryContestFall.fall_from_one.

Here $G(x) = 1 - (1 - \frac{4x}{3})^2$ on $[0, \frac{3}{4}]$, $F(x) = pG(x) + (1 - p)G(x - \frac{1}{4})$, $W_1 = F(1 - G)$ and $W_2 = FG(1 - G)$. By M16, integrating the piecewise polynomials on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{3}{4}]$ and $[\frac{3}{4}, 1]$,

$$\frac{D(1)}{V} = -\frac{11p^2}{162} + \frac{1901p}{21870} - \frac{208}{10935}, \quad \frac{D(2)}{V} = -\frac{4933p^2}{393660} + \frac{77219p}{2755620} - \frac{10672}{688905},$$

so

$$p = \frac{1}{2} : \frac{D(1)}{V} = \frac{653}{87480}, \quad \frac{D(2)}{V} = -\frac{10169}{2204496}, \quad p = \frac{1}{4} : \frac{D(1)}{V} = -\frac{179}{116640}.$$

□

Proof of M21

Lean: EntryContestMustFall.must_fall_general, EntryContestMustFall.shift_dominates, EntryContestMustFall.stepQ_bern, EntryContestMustFall.must_fall, EntryContestMustFall.witness_must_fall, EntryContestMustFall.unif_shiftMono, EntryContestMustFall.uniform_never_gains.

With $F = pG + (1 - p)G_t$ and $W_Q = G^{Q-1}W_1$, M16 gives

$$D(Q) = -V(1 - p) \left(\int W_Q dG_t - \int W_Q dG \right).$$

G_t has no mass on $(-\infty, t]$, $dG_t \geq dG$ on (t, ∞) by M19, and $W_1 \geq \eta = C(a)(1 - G(b)) > 0$ on $(a, b]$. Splitting at t and a ,

$$\int W_Q dG_t - \int W_Q dG \geq G(a)^{Q-1}A - G(t)^{Q-1}B,$$

$$A = \int_{(a, \infty)} W_1 d(G_t - G) \geq \eta (G_t - G)((a, b]) > 0, \quad B = \int_{(-\infty, t]} W_1 dG \geq 0.$$

With $r_0 = G(t)/G(a) < 1$ and $r_0^{Q_0-1}(B+1) < A$, the bound is positive for $Q \geq Q_0$. For M20's base law, $t = \frac{1}{4}$ and

$$G((\frac{1}{2}, \frac{5}{8}]) = \frac{35}{36} - \frac{8}{9} = \frac{1}{12} < \frac{7}{36} = \frac{3}{4} - \frac{5}{9} = G_t((\frac{1}{2}, \frac{5}{8}]).$$

A uniform base score is shift-monotone, and M18 gives $D(Q) \geq 0$, so it never meets the condition. □

Proof of M22

Lean: `EntryContestPMUWitness.pairA_fosd, EntryContestPMUWitness.pairB_fosd,`
`EntryContestPMUWitness.pairA_step_one, EntryContestPMUWitness.pairA_step_two,`
`EntryContestPMUWitness.pairB_step, EntryContestPMUWitness.fosd_does_not_sign.`

Pair one, with $y = 1 - x$, $W_1 = x(1 - x)^2$, $W_2 = x(1 - (1 - x)^2)(1 - x)^2$ and $dG = 2(1 - x)dx$:

$$\int W_1 dF = \frac{1}{12}, \quad \int W_1 dG = \frac{1}{10}, \quad \int W_2 dF = \int_0^1 (y^2 - y^3 - y^4 + y^5) dy = \frac{1}{20}, \quad \int W_2 dG = \frac{11}{210},$$

$$\frac{D(1)}{V} = \frac{1}{10} - \frac{1}{12} = \frac{1}{60}, \quad \frac{D(2)}{V} = \frac{11}{210} - \frac{1}{20} = \frac{1}{420}.$$

Pair two, $W_Q = x^{Q+1} - x^{Q+2}$ and $dF = 2x dx$:

$$\int W_Q dF = \frac{2}{(Q+3)(Q+4)}, \quad \int W_Q dG = \frac{1}{(Q+2)(Q+3)}, \quad \frac{D(Q)}{V} = -\frac{2(Q+2) - (Q+4)}{(Q+2)(Q+3)(Q+4)}.$$

□

B Evidence and verification

C Terminology

D Primitives and scope

E Refuted conjectures

F Responses to the referee reports

References